# Linear Algebra — Model Test 1: Answers and Methods

Covers Sections 1.1–1.3 and 2.1–2.6 · Level: standard (same as the sample test)

## Key methods used in this test

- **Dot product and length (1.2):** $v \cdot w = \sum v_i w_i$, $\|v\| = \sqrt{v \cdot v}$, $u = v/\|v\|$, $\cos\theta = \dfrac{v \cdot w}{\|v\|\|w\|}$; perpendicular $\iff v \cdot w = 0$.
- **Matrix product (1.3, 2.4):** $(AB)_{ij}$ = row $i$ of $A$ · column $j$ of $B$. Associative, so multiply neighbours first.
- **Polynomial of a matrix:** compute $A^2, A^3$ first; the constant $c$ becomes $cI$.
- **Elimination (2.2):** multiplier $\ell_{ij} = \dfrac{\text{entry to eliminate}}{\text{pivot}}$, then $R_i - \ell_{ij} R_j$. Reach upper triangular $U$, then back substitution.
- **Inverse (2.5):** row-reduce $(A \,|\, I)$ to $(I \,|\, A^{-1})$.
- $A = LU$ **(2.6):** $U$ is the result of elimination; $L$ holds the multipliers $\ell_{ij}$ below its diagonal of 1's.

## 1 Vectors

**Method.** Work component by component. Use $\|v\| = \sqrt{v \cdot v}$ and $\cos\theta = \frac{v \cdot w}{\|v\|\|w\|}$.

**(a)** $2v - 3w = (2,4,4) - (9,-3,3) = (-7,7,1)$, and $v \cdot w = 1 \cdot 3 + 2 \cdot (-1) + 2 \cdot 1 = 3$.
**(b)** $\|v\| = \sqrt{1+4+4} = 3$, so $u = \frac{1}{3}v = \left(\frac{1}{3}, \frac{2}{3}, \frac{2}{3}\right)$ (check: $\|u\|^2 = \frac{1+4+4}{9} = 1$).
**(c)** $a \cdot b = 1$, $\|a\| = \|b\| = \sqrt{2}$, so $\cos\theta = \frac{1}{2}$ and $\theta = 60° = \frac{\pi}{3}$.
**(d)** $(1, c, 2) \cdot (3, -1, 1) = 3 - c + 2 = 0 \Rightarrow c = 5$.

**Answer.** (a) $(-7,7,1)$, $v \cdot w = 3$; (b) $\|v\| = 3$, $u = (\frac{1}{3}, \frac{2}{3}, \frac{2}{3})$; (c) 60°; (d) $c = 5$.

## 2 Matrix operations

**(1)**

**Method.** Multiply $AB$ first ($2 \times 4$ times $4 \times 3$ gives $2 \times 3$), then multiply by the $3 \times 2$ matrix.

Entry $(1,1)$ of $AB$: $2 \cdot 1 + (-1)(-1) + 1 \cdot 0 + 0 \cdot 3 = 3$. All entries:

$$AB = \begin{pmatrix} 3 & -1 & 2 \\ 1 & 2 & 7 \end{pmatrix}, \qquad ABC = \begin{pmatrix} 3 & -1 & 2 \\ 1 & 2 & 7 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 2 & 0 \\ -3 & 1 \end{pmatrix} = \begin{pmatrix} -5 & -1 \\ -16 & 6 \end{pmatrix}.$$

**Answer.** $ABC = \begin{pmatrix} -5 & -1 \\ -16 & 6 \end{pmatrix}$

**(2)**

**Method.** Compute $A^2$, $A^3$, then $f(A) = A^3 - A^2 + 2A + I$.

$$A^2 = \begin{pmatrix} -3 & 4 \\ -4 & -3 \end{pmatrix}, \qquad A^3 = A^2 A = \begin{pmatrix} -11 & -2 \\ 2 & -11 \end{pmatrix}.$$

$$f(A) = \begin{pmatrix} -11 & -2 \\ 2 & -11 \end{pmatrix} - \begin{pmatrix} -3 & 4 \\ -4 & -3 \end{pmatrix} + 2\begin{pmatrix} 1 & 2 \\ -2 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} -5 & -2 \\ 2 & -5 \end{pmatrix}.$$

**Answer.** $f(A) = \begin{pmatrix} -5 & -2 \\ 2 & -5 \end{pmatrix}$

## 3 Elimination

**Method.** Eliminate below each pivot (no row exchanges needed), then back-substitute from the last equation upward.

$$\left(\begin{array}{ccc|c} 2 & 1 & 1 & 5 \\ 4 & -6 & 0 & -2 \\ -2 & 7 & 2 & 9 \end{array}\right)$$

$$\xrightarrow{R_2-2R_1,\ R_3+R_1} \left(\begin{array}{ccc|c} 2 & 1 & 1 & 5 \\ 0 & -8 & -2 & -12 \\ 0 & 8 & 3 & 14 \end{array}\right)$$

$$\xrightarrow{R_3+R_2} \left(\begin{array}{ccc|c} 2 & 1 & 1 & 5 \\ 0 & -8 & -2 & -12 \\ 0 & 0 & 1 & 2 \end{array}\right)$$

Multipliers: $\ell_{21} = 2$, $\ell_{31} = -1$, $\ell_{32} = -1$. Pivots: $2$, $-8$, $1$.
Back substitution: $z = 2$; $-8y - 2z = -12 \Rightarrow -8y = -8 \Rightarrow y = 1$; $2x + y + z = 5 \Rightarrow 2x = 2 \Rightarrow x = 1$.

**Answer.** $(x, y, z) = (1, 1, 2)$; pivots $2, -8, 1$.

## 4 Inverse matrix

**Method.** Row-reduce $(A \,|\, I_3)$ until the left block is $I_3$; the right block is $A^{-1}$.

$$\left(\begin{array}{ccc|ccc} 1 & 0 & -1 & 1 & 0 & 0 \\ -1 & -1 & 1 & 0 & 1 & 0 \\ 1 & -2 & -2 & 0 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_2+R_1,\ R_3-R_1} \left(\begin{array}{ccc|ccc} 1 & 0 & -1 & 1 & 0 & 0 \\ 0 & -1 & 0 & 1 & 1 & 0 \\ 0 & -2 & -1 & -1 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{-R_2,\ R_3+2R_2} \left(\begin{array}{ccc|ccc} 1 & 0 & -1 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & -1 & 0 \\ 0 & 0 & -1 & -3 & -2 & 1 \end{array}\right)$$

$$\xrightarrow{-R_3,\, R_1+R_3} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 4 & 2 & -1 \\ 0 & 1 & 0 & -1 & -1 & 0 \\ 0 & 0 & 1 & 3 & 2 & -1 \end{array}\right)$$

**Answer.** $A^{-1} = \begin{pmatrix} 4 & 2 & -1 \\ -1 & -1 & 0 \\ 3 & 2 & -1 \end{pmatrix}$ (check: $AA^{-1} = I$).

## 5 $A = LU$

**Method.** Do elimination without row exchanges and record each multiplier $\ell_{ij}$; put it in position $(i, j)$ of $L$.

$$\begin{pmatrix} 1 & 2 & 1 \\ 2 & 5 & 4 \\ 3 & 8 & 9 \end{pmatrix}$$

$$\xrightarrow{R_2-2R_1,\, R_3-3R_1} \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \\ 0 & 2 & 6 \end{pmatrix}$$

$$\xrightarrow{R_3-2R_2} \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{pmatrix}$$

Multipliers: $\ell_{21} = 2$, $\ell_{31} = 3$, $\ell_{32} = 2$.

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{pmatrix}, \qquad U = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{pmatrix}.$$

Check: $LU = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 5 & 4 \\ 3 & 8 & 9 \end{pmatrix} = A$ ✓

**Answer.** $L = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{pmatrix}$, $U = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{pmatrix}$